AKSHIT SRIVASTAVA

+91-8104394240 | as56intn@gmail.com | LinkedIn | GitHub

EDUCATION

Birla Institute of Technology, Mesra Aug 2023 – Present
Bachelor of Technology in Computer Science & Engineering

EXPERIENCE

1. Research Intern – IIIT Nagpur May – Jul 2026
Machine learning for remote sensing / geospatial AI — Python, PyTorch, Scikit-learn

- Worked on soil classification and mapping from hyperspectral cubes, and estimation of soil properties such as organic carbon, moisture, nitrogen, pH, and salinity from spectra.
- Applied dimensionality reduction and band selection (PCA, feature selection) and trained deep learning models on spectra (1D/3D CNNs, Transformers) for small-data settings, using leakage-free, cube-level evaluation.
- The research proposing PG-SD-SVM and a explainable Transformer-based spectral prediction (99.12% accuracy, 99.04% Macro-F1).

2. Machine Learning Intern – Infosys Springboard Feb – Apr 2026
TransferIQ — Python, FastAPI, XGBoost, LSTM

- Built and shipped an end-to-end prediction pipeline from scratch, combining LSTM-derived features with XGBoost for a football transfer-value model.
- Designed preprocessing, feature engineering, model training, and inference workflows as modular, reusable components.
- Exposed model inference through a FastAPI service and delivered a deployable production-ready application.

3. AI Intern – Edunet Foundation Jun – Jul 2025
Carbon Emission Prediction — Python, Pandas, Scikit-learn

- Built an end-to-end ML pipeline for CO2 emission prediction using environmental and climate-related data.
- Applied preprocessing, feature scaling, cross-validation, and ensemble regression for predictive modeling.

PROJECTS

1. CodePilot AI – AI Code Review Platform GitHub | Live
Built in a team of three, a web app that reviews code in 12 languages, flagging each issue by severity and category (bug, security, performance, and more) with a suggested fix and a rewritten file. Code is syntax-checked in the browser with Tree-sitter (WebAssembly) before review; a TypeScript Cloudflare Worker verifies Clerk sessions, calls a self-hosted Qwen LLM with a Dockerised fallback model, and stores review history in Supabase PostgreSQL. Evaluated on the QuixBugs and Juliet benchmarks, with CI/CD through GitHub Actions.
Tech: React, TypeScript, Hono, Cloudflare Workers, Supabase, Clerk, Tree-sitter, Docker, GitHub Actions.

2. AI Career Mentor GitHub | Live
Built an AI-powered career platform from scratch that evaluates resumes, grades role fit, identifies skill gaps, and generates personalized learning roadmaps. Integrated Groq-hosted LLM inference with a native Python Cloudflare Worker to generate structured, role-specific recommendations, and implemented persistent assessment history and roadmap tracking using Cloudflare D1.
Tech: Python, React, Groq, Cloudflare Workers, D1.

SKILLS

Languages: Python, SQL, JavaScript, TypeScript
Frontend: React, Vite
Backend & APIs: FastAPI, Flask, REST APIs, Cloudflare Workers
Machine Learning: PyTorch, Scikit-learn, XGBoost, LightGBM, SVM, LSTM, CNNs, PCA, SHAP
AI & LLM Integration: Groq, Qwen, LLM Integration, NLP, VADER, TextBlob
Data & Scientific: NumPy, Pandas, Matplotlib, SciPy, Scikit-image
Databases: PostgreSQL, MySQL, Supabase, Cloudflare D1
Tools & Deployment: Git, GitHub Actions, Docker, CI & CD, Linux, MLflow, Jupyter

CERTIFICATIONS & RECOMMENDATIONS

- Research Internship Certificate – Indian Institute of Information Technology, Nagpur
- Letter of Recommendation – Indian Institute of Information Technology, Nagpur
- Machine Learning Internship Certificate – Infosys Springboard 6.0